

M13 Batch 1 — Full Set

M13.1 Theory of Flight · Rotary wing · PDF pages 1–63

Terminology and blade geometry, lift and gyroscopic precession, classification of rotary-wing aircraft, the cyclic, collective and anti-torque controls, swash plate and flapping, ground effect and autorotation.

160 questions · Level 3: 59 · Level 2: 57 · Level 1: 44

● marks a question built on a fact your instructor highlighted or underlined in the manual (150 of 160). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Terminology and blade geometry

PDF pages 4–20 · 35 questions

LEVEL 1 — RECALL

1. What is the primary airfoil of the helicopter ●
 - a. The tail rotor
 - b. The fuselage
 - c. The main rotor
2. What is an airfoil by definition ●
 - a. Any surface which generates a useful static pressure of the air
 - b. Any surface which generates a useful dynamic reaction of the air
 - c. Only a wing surface which generates lift in forward flight
3. Which two forces are named as the useful reaction of the air on an airfoil ●
 - a. Thrust and weight
 - b. Lift and thrust
 - c. Lift and drag
4. What is the leading edge of a rotor blade ●
 - a. The edge that cuts into the wind when the rotor is turning
 - b. The outer end of the blade opposite the root
 - c. The edge opposite the one that cuts into the wind
5. What is the trailing edge of a rotor blade ●
 - a. The edge opposite the leading edge
 - b. The end attached to the rotor mast
 - c. The edge that cuts into the wind
6. What is the advancing blade ●
 - a. The blade at the tail boom
 - b. The backward moving blade
 - c. The forward moving blade
7. What is the retreating blade ●
 - a. The backward moving blade
 - b. The forward moving blade
 - c. The blade over the cockpit

8. What is the relative wind ●
 - a. The direction of the airflow with respect to the ground
 - b. The speed of the natural wind at the landing site
 - c. The direction of the airflow with respect to the rotor blade
9. How is the relative wind directed in relation to the flight path of the blade ●
 - a. Always perpendicular to the path
 - b. Always opposite to the flight path
 - c. Always along the same flight path
10. What is the axis of rotation ●
 - a. The axis around which the rotor blades are turning
 - b. The plane formed by the average tip path
 - c. The line from the leading edge to the trailing edge
11. What forms the plane of rotation ●
 - a. The chord line of the blades
 - b. The average tip path of the blades
 - c. The flapping hinges of the blades
12. What is the pitch angle ●
 - a. The acute angle between the chord and the relative wind
 - b. The acute angle between the axis and the plane of rotation
 - c. The acute angle between the chord and the plane of rotation
13. Which cockpit control changes the pitch angle of all main rotor blades ●
 - a. The tail rotor pedals
 - b. The collective pitch control
 - c. The cyclic pitch stick
14. When is the angle of attack equal to the pitch angle ●
 - a. When the relative wind is parallel to the axis of rotation
 - b. When the relative wind is in the same direction as the plane of rotation
 - c. When the relative wind is perpendicular to the plane of rotation
15. The pilot increases the pitch angle. What happens to the angle of attack ●
 - a. It is decreased
 - b. It remains unchanged
 - c. It is also increased

LEVEL 2 — UNDERSTAND AND EXPLAIN

16. State why a technician has to study the basic principles of helicopter flight ●
 - a. To calculate the lift and thrust produced by the main rotor blades
 - b. To obtain the flight crew licence required for ground running of the rotor
 - c. To understand the maintenance and troubleshooting practices on the systems
17. Compare the airfoils of a fixed-wing aircraft with those of a helicopter ●
 - a. Primarily the propellers on the fixed-wing aircraft, the fuselage on the helicopter
 - b. Primarily the wings on the fixed-wing aircraft, the main rotor on the helicopter
 - c. Primarily the fuselage on the fixed-wing aircraft, the tail rotor on the helicopter

18. Why is the helicopter often referred to as a rotary-wing aircraft ●
 - a. Its primary airfoil is the tail rotor
 - b. Its primary airfoil is the main rotor
 - c. Its fuselage acts as a rotating wing
19. The main rotor blades are the airfoils of a helicopter. Which of the two forces they produce holds the aircraft up against gravity? ●
 - a. Lift
 - b. Thrust
 - c. Drag
20. A rotor blade is moving forward and downward through the air. In which direction does its relative wind act? ●
 - a. Forward and upward
 - b. Backward and upward
 - c. Backward and downward
21. Explain why the relative wind on a rotor blade changes when a hovering helicopter begins to move horizontally ●
 - a. The horizontal movement of the helicopter changes the chord line of the rotor blade
 - b. The horizontal movement of the helicopter changes the span of the rotor blade
 - c. Movement of the helicopter through the air is one of the factors that affect it
22. A helicopter stands with its rotor stopped while wind blows over the blades. Which statement about the relative wind is correct? ●
 - a. There is no relative wind because the rotor is stopped
 - b. The relative wind is created only by blade flapping
 - c. The wind blowing over the blade creates the relative wind
23. A helicopter is flying and the blades flap during each revolution, so each tip follows a slightly different path. Which path defines the plane of rotation? ●
 - a. The average tip path of the blades
 - b. The path of the blade roots
 - c. The highest tip path of the blades
24. Explain how one movement of the collective pitch control is able to change both the pitch angle and the angle of attack ●
 - a. It lifts the whole blade, moving the axis of rotation against the chord
 - b. It tilts the plane of rotation, moving the relative wind against the chord
 - c. It rotates the blade around the hub axis, moving the chord against each reference
25. Which statement correctly distinguishes pitch angle from angle of attack? ●
 - a. Pitch angle is referenced to the relative wind, angle of attack to the plane of rotation
 - b. Pitch angle is referenced to the axis of rotation, angle of attack to the span line
 - c. Pitch angle is referenced to the plane of rotation, angle of attack to the relative wind
26. Why do the same factors that affect the relative wind also affect the angle of attack ●
 - a. Because the angle of attack depends upon the relative wind
 - b. Because the relative wind depends upon the pitch angle
 - c. Because the angle of attack depends upon the blade span

27. The pilot holds the pitch angle constant, but outside influences change the relative wind. What happens to the angle of attack?
- It stays constant, because the pitch is constant
 - It changes, because it depends on the relative wind
 - It becomes equal to the pitch angle of the blade

LEVEL 3 — APPLY AND REASON

28. Two reference dimensions of a rotor blade are quoted, one measured along the blade centreline and one across the blade section. Which pair is correct? ●
- Span along the centreline, root across the section
 - Span along the centreline, chord across the section
 - Chord along the centreline, span across the section
29. A helicopter is in forward flight and one blade is on the advancing side. From which direction does the relative wind reach that blade? ●
- Horizontally forward, with the blade movement
 - Horizontally backward, opposite to the blade movement
 - Backward and upward, across the blade movement
30. A blade is moving forward and at the same time flapping upward. Applying the rule for relative wind, from which direction does the relative wind come? ●
- Forward and downward
 - Backward and upward
 - Backward and downward
31. State what creates the relative wind on a helicopter hovering in completely still air ●
- The forward speed of the helicopter
 - The turning of the rotor blades
 - The wind speed and the wind direction
32. A blade is set so that its chord lies exactly along the average tip path of the blades. What is its pitch angle? ●
- Ninety degrees, because the chord and the axis of rotation coincide
 - Equal to the angle of attack, since the relative wind coincides
 - Zero, because the chord and the plane of rotation coincide
33. On a turning blade the pitch angle and the angle of attack are found to be unequal. What does that tell you? ●
- The blade is not fitted with a feathering hinge at its root
 - The relative wind is not in the direction of the plane of rotation
 - The collective pitch control has not been moved from its stop
34. The relative wind on a blade lies in the same direction as the plane of rotation and the pilot then increases the pitch angle. What happens to the angle of attack? ●
- It stays constant while the pitch angle increases
 - It increases and stays equal to the pitch angle
 - It increases and becomes larger than the pitch angle

35. Give the explanation for the angle of attack of a blade changing while the collective pitch control is held still ●
- An outside influence has changed the chord of the blade
 - An outside influence has changed the plane of rotation
 - An outside influence has changed the relative wind

Answers — Terminology and blade geometry

- c The manual names the main rotor as the helicopter's primary airfoil. (L1, PDF p. 4)
- b The manual defines an airfoil as any surface giving a useful dynamic reaction of the air. (L1, PDF p. 6)
- b The text names lift and thrust as the two forces needed for flying and manoeuvring. (L1, PDF p. 6)
- a The leading edge cuts into the wind. (L1, PDF p. 8)
- a The trailing edge is opposite the leading edge. (L1, PDF p. 8)
- c The blade moving forward is the advancing blade. (L1, PDF p. 10)
- a The blade moving backward is the retreating blade. (L1, PDF p. 10)
- c Relative wind is airflow direction relative to the blade. (L1, PDF p. 12)
- b The relative wind is always opposite to the blade's flight path. (L1, PDF p. 12)
- a The rotor blades turn around the axis of rotation. (L1, PDF p. 16)
- b The plane of rotation is formed by the average blade tip path. (L1, PDF p. 16)
- c Pitch angle is measured from the plane of rotation to the chord. (L1, PDF p. 18)
- b The collective pitch control changes the pitch angle. (L1, PDF p. 18)
- b If the relative wind lies in the plane of rotation, both angles use the same reference line. (L1, PDF p. 20)
- c Pitch and angle of attack change together. (L1, PDF p. 20)
- c The manual gives understanding of maintenance and troubleshooting as the reason for studying the principles. (L2, PDF p. 4)
- b The wings are the primary airfoil of a fixed-wing aircraft and the main rotor is the primary airfoil of the helicopter. (L2, PDF p. 4)
- b Because its main lifting surface is a rotating rotor, the helicopter is called rotary-wing. (L2, PDF p. 4)
- a Of the two forces the airfoil produces, lift is the one the manual gives as counteracting gravity. (L2, PDF p. 6)
- b The relative wind is always exactly opposite to the flight path of the blade, so forward and downward movement gives a backward and upward relative wind. (L2, PDF p. 12)
- c Horizontal movement of the helicopter is one of the factors that affect the relative wind. (L2, PDF p. 14)
- c Even with the rotor stopped, wind over the blade is airflow relative to the blade. (L2, PDF p. 14)
- a The manual defines it by the average tip path. (L2, PDF p. 16)
- c Both angles are measured from the chord, so rotating the blade about the hub axis changes both. (L2, PDF p. 18)
- c Both use the chord, but pitch angle uses the plane of rotation and angle of attack uses the relative wind. (L2, PDF p. 18)
- a Angle of attack is measured to the relative wind, so what changes one changes the other. (L2, PDF p. 20)
- b Angle of attack is also affected by relative wind changes, not only by pitch. (L2, PDF p. 20)
- b Span runs root to tip along the centreline, and the chord runs leading edge to trailing edge across the section. (L3, PDF p. 6)
- b The advancing blade moves forward, and the relative wind is always opposite to the blade's flight path. (L3, PDF p. 10)
- c The relative wind moves exactly opposite to the blade path, so forward-and-upward blade motion gives a backward-and-downward relative wind. (L3, PDF p. 12)
- b With no wind and no forward speed, only the turning of the blades is left to produce the relative wind. (L3, PDF p. 14)
- c The pitch angle is measured between the chord and the plane of rotation, so a chord lying in that plane gives zero. (L3, PDF p. 16)

- 33. b** The two angles are only equal when the relative wind lies in the plane of rotation, so unequal angles mean it does not. (L3, PDF p. 18)
- 34. b** With the relative wind in the plane of rotation the two angles are equal, and the angle of attack follows the pitch angle up. (L3, PDF p. 20)
- 35. c** Angle of attack is measured from the relative wind, so it changes when outside influences move the relative wind. (L3, PDF p. 20)

Lift and gyroscopic precession

PDF pages 22–24 • 13 questions

LEVEL 1 — RECALL

36. What is lift ●
- The force produced by torque reaction
 - The force that is produced by the airfoil
 - The force that is produced by gravity
37. In which direction does lift act in relation to the relative wind ●
- Parallel to the relative wind
 - Opposite to the relative wind
 - Perpendicular to the relative wind
38. To which bodies does gyroscopic precession apply ●
- All rotating bodies
 - Only the tail rotor
 - Only the main rotor
39. How is gyroscopic precession defined ●
- The change of the rotational axis as a result of an external force
 - The change of the rotor RPM as a result of an external force
 - The turning of the fuselage opposite to the rotor

LEVEL 2 — UNDERSTAND AND EXPLAIN

40. Explain why gyroscopic precession has to be considered on the tail rotor as well as on the main rotor ●
- Because the tail rotor blades have feathering hinges
 - Because the tail rotor turns opposite to the fuselage
 - Because the effect applies to every rotating body
41. In the example wheel, an external force tries to turn the wheel around the input axis. Around which axis does the movement actually take place? ●
- Only around the input axis
 - Only around the spin axis
 - Around the output axis
42. Why must the steering input to the rotor be applied 90° before the point where the tilt is wanted ●
- Because the swash plate is mounted 90° to the rotor
 - Because the output occurs 90° earlier in the direction of rotation
 - Because the output occurs 90° later in the direction of rotation
43. A downward force is applied on the right side of a counterclockwise turning rotor. Where does the output appear? ●
- Right in front
 - Right at the rear
 - On the left side

LEVEL 3 — APPLY AND REASON

44. State what happens to the direction in which lift acts on a blade when the relative wind changes direction ●
- It changes with it, staying perpendicular to the relative wind
 - It changes with it, staying parallel to the relative wind
 - It stays fixed, remaining perpendicular to the plane of rotation
45. State what must happen to the plane of rotation when gyroscopic precession tilts the axis of rotation ●
- It stays level, while the axis tilts away from it
 - It tilts with the axis, staying perpendicular to it
 - It tilts the opposite way, staying parallel to the axis
46. A helicopter rotor turns counterclockwise. The tip-path plane must tilt backward for rearward flight. Following the precession rule, where must the downward force be put in? ●
- On the left side
 - Right at the rear
 - On the right side
47. A helicopter rotor turns counterclockwise. The tip-path plane must tilt to the left. Following the precession rule, where must the downward force be put in? ●
- On the left side
 - Right in front
 - Right at the rear
48. A downward force is applied at the rear of a counterclockwise turning rotor. Where does the tip-path plane tilt down? ●
- On the right side
 - Right in front
 - On the left side

Answers — Lift and gyroscopic precession

36. **b** Lift is the airfoil's force. (L1, PDF p. 22)
37. **c** Lift is perpendicular to the relative wind. (L1, PDF p. 22)
38. **a** It applies to all rotating bodies, including both rotors. (L1, PDF p. 24)
39. **a** Precession is a change of the rotational axis caused by an external force. (L1, PDF p. 24)
40. **c** The principle is stated to apply to all rotating bodies, which includes the tail rotor. (L2, PDF p. 24)
41. **c** Because of precession the movement occurs around the output axis. (L2, PDF p. 24)
42. **c** Precession shifts the result 90° in the direction of rotation, so the input must lead by 90°. (L2, PDF p. 24)
43. **a** 90° later in counterclockwise rotation from the right side is the front. (L2, PDF p. 24)
44. **a** Lift is defined as perpendicular to the relative wind, so its direction follows the relative wind. (L3, PDF p. 22)
45. **b** The plane of rotation is always perpendicular to the axis of rotation, so it tilts with the axis. (L3, PDF p. 24)
46. **a** The output is wanted at the rear; 90° earlier in counterclockwise rotation is the left side. (L3, PDF p. 24)
47. **b** The output is wanted on the left; 90° earlier in counterclockwise rotation is the front. (L3, PDF p. 24)
48. **a** 90° later than the rear in counterclockwise rotation is the right side. (L3, PDF p. 24)

Classification of rotary-wing aircraft

PDF pages 26–34 • 10 questions

LEVEL 1 — RECALL

49. Which law explains torque reaction on a helicopter fuselage ●
- a. Newton's Third Law
 - b. Newton's law of inertia
 - c. Newton's Second Law
50. In which direction does the helicopter fuselage tend to rotate due to torque ●
- a. Around its lateral axis, nose down
 - b. In the same direction as the rotor blades
 - c. Opposite to the direction of the rotor blades

LEVEL 2 — UNDERSTAND AND EXPLAIN

51. Explain why the torque of the main rotor must be counteracted before a flight is possible ●
- a. Because the fuselage would otherwise turn in the direction opposite to the rotor blades
 - b. Because the main rotor would otherwise produce thrust instead of lift
 - c. Because the rotor blades would otherwise turn in the direction opposite to the fuselage
52. A helicopter's main rotor turns counterclockwise seen from above and the tail rotor is not yet producing thrust. Which way does the fuselage tend to turn? ●
- a. Not at all
 - b. Clockwise
 - c. Counterclockwise

LEVEL 3 — APPLY AND REASON

53. A pilot raises the collective lever and the turning force on the fuselage rises. Why must the tail rotor thrust then change too? ●
- a. Because it is what balances the lift of the main rotor
 - b. Because it is what balances the drag of the main rotor
 - c. Because it is what compensates the torque on the airframe
54. Name what a gyrocopter in normal flight and a helicopter after an engine failure have in common
- a. The rotor is held at a fixed pitch angle by the collective
 - b. The rotor is driven by the airflow in autorotation
 - c. The rotor is driven by the engine through the drive shafts
55. The lift-producing rotor of a gyroplane stays connected to the engine in flight. What follows from that?
- a. The rotor torque has to be compensated
 - b. The rotor can no longer stall in flight
 - c. The rotor has to be tilted slightly aft
56. Compare how the rotor of a gyrocopter and the rotor of a gyroplane receive their power in flight
- a. The gyroplane rotor is driven by the airflow, the gyrocopter rotor shares the engine power
 - b. The gyrocopter rotor is driven by the airflow, the gyroplane rotor shares the engine power
 - c. Both rotors are driven by the airflow and the propellers share the engine power

- 57.** A convertiplane has accelerated to the airspeed at which its rotor turns freely in autorotation. What is then providing lift and what is providing thrust?
- The wings give the lift, the rotor gives the thrust
 - The rotor gives the lift, the propellers or turbines give the thrust
 - The wings give the lift, the propellers or turbines give the thrust
- 58.** Compare the rotor of a convertiplane in cruise with the rotor of a type 3 transformation helicopter in cruise
- The type 3 rotor turns in autorotation, the convertiplane rotor is shut down and stowed
 - The convertiplane rotor turns in autorotation, the type 3 rotor is shut down and stowed
 - Both rotors turn in autorotation and carry part of the aircraft weight

Answers — Classification of rotary-wing aircraft

- 49. a** Torque reaction is action and reaction, Newton's Third Law. (L1, PDF p. 26)
- 50. c** Action and reaction: the fuselage turns opposite to the rotor. (L1, PDF p. 26)
- 51. a** Torque makes the fuselage rotate opposite to the blades, so it must be counteracted before flight. (L2, PDF p. 26)
- 52. b** The fuselage reaction is opposite to a counterclockwise rotor, so clockwise. (L2, PDF p. 26)
- 53. c** The tail rotor is what compensates the torque, so a change in torque has to be matched by a change in its thrust. (L3, PDF p. 26)
- 54. b** Both have an unpowered rotor kept turning by the airflow, which is autorotation. (L3, PDF p. 28)
- 55. a** An engine-driven rotor produces a torque reaction on the airframe, so it must be compensated. (L3, PDF p. 30)
- 56. b** The gyrocopter rotor is unpowered in flight while the gyroplane shares engine power between rotor and propellers. (L3, PDF p. 30)
- 57. c** Once the rotor autorotates it makes no lift, so the wings carry the aircraft and the propellers or turbines push it. (L3, PDF p. 32)
- 58. b** The convertiplane rotor keeps turning in autorotation while the type 3 rotor is deactivated and stowed. (L3, PDF p. 34)

Cyclic, collective and anti-torque controls

PDF pages 36–42 • 29 questions

LEVEL 1 — RECALL

59. What does lift counteract according to the helicopter description ●
- a. Thrust
 - b. Torque
 - c. Weight
60. After an engine failure or broken drive shafts, how can a helicopter land safely ●
- a. By using translating tendency
 - b. By performing an autorotation
 - c. By using ground effect
61. How do helicopters change direction, unlike other aircraft ●
- a. By changing the angle of the rotor blade plane
 - b. By changing the RPM of the main rotor
 - c. By deflecting control surfaces on the fuselage
62. The pilot moves the cyclic stick forward. What does the rotor plane do ●
- a. It tilts backward
 - b. It tilts forward
 - c. It moves upward
63. What does the pilot do to the main rotor blades by moving the collective lever upwards ●
- a. Increases the pitch angle of all blades at the same time
 - b. Increases the pitch angle of each blade once per revolution
 - c. Tilts the rotor plane in the direction of flight
64. The pilot moves the collective lever upwards. What does the helicopter do ●
- a. It descends
 - b. It climbs
 - c. It turns left
65. Which control does the pilot use to turn the helicopter around the rotor mast axis ●
- a. The collective lever
 - b. The cyclic pitch stick
 - c. The tail rotor pedals
66. What do the pedals change ●
- a. The pitch angle of the tail rotor blades
 - b. The pitch angle of the main rotor blades
 - c. The tilt of the main rotor plane

LEVEL 2 — UNDERSTAND AND EXPLAIN

67. State what the helicopter gains from being able to tilt its rotor plane slightly in any direction ●
- a. Thrust can act in all directions, giving directional control
 - b. Torque can act in all directions, giving heading control
 - c. Lift can act in all directions, giving collective control

- 68.** Why does a helicopter need no pull or push propellers
- Because the rotors generate both lift and thrust
 - Because the tail rotor generates the thrust
 - Because the rotors are driven by the airflow
- 69.** How do the rotors of a helicopter additionally ensure directional control ●
- The rotor plane may be tilted so thrust can act in all directions
 - The rotor RPM is changed so thrust acts in all directions
 - Control surfaces on the rotor blades are deflected
- 70.** During an autorotation after an engine failure, which statement about control is correct? ●
- The helicopter is not controllable and lands wherever the descent takes it
 - The helicopter is fully controllable and the pilot can choose the touchdown point
 - Only the pedals remain effective and the touchdown point cannot be chosen
- 71.** Why must the rotor plane be tilted when changing from hover to forward flight ●
- Because the rotor RPM must be increased for forward flight
 - Because the tail rotor thrust acts in the direction of flight
 - Because rotor thrust always acts perpendicular to the rotor plane
- 72.** How does cyclic control create different amounts of lift within the rotor plane ●
- The rotor RPM is increased and decreased during each revolution
 - All blades change their pitch angle at the same time together
 - The blade changes its angle of incidence during each revolution
- 73.** Explain why moving the collective lever also alters the torque effect on the fuselage ●
- Because the movement of the lever also changes the tail rotor pitch
 - Because the movement of the lever also changes the engine power
 - Because the movement of the lever also tilts the rotor plane
- 74.** Besides changing the blade pitch, which other effect does a movement of the collective lever have? ●
- It changes engine power and causes effects like torque
 - It tilts the rotor plane and causes translating tendency
 - It changes tail rotor pitch and causes precession
- 75.** A pilot wants to yaw the helicopter around the rotor mast axis. How do the pedals bring this about? ●
- They change the pitch angle of the main rotor blades and so vary its thrust
 - They change the pitch angle of the tail rotor blades and so vary its thrust
 - They tilt the tail rotor plane and so vary the direction of its thrust
- 76.** How do the pedals vary the tail rotor thrust ●
- By changing the pitch angle of the tail rotor blades
 - By changing the RPM of the tail rotor drive shaft
 - By tilting the plane of rotation of the tail rotor

LEVEL 3 — APPLY AND REASON

- 77.** State the penalty that follows from a helicopter rotor generating both lift and thrust by itself ●
- Its rotor cannot be tilted out of the horizontal plane
 - Its torque reaction on the fuselage has to be counteracted
 - Its downwash cannot escape from under the fuselage

- 78.** Explain how a helicopter is able to fly sideways ●
- The rotor plane is tilted sideways, so the thrust acts sideways
 - The tail rotor thrust is increased, so the fuselage is pushed sideways
 - The blade pitch is increased on one side, so the lift acts sideways
- 79.** Compare how the cyclic control and the collective control change the pitch angle of the main rotor blades ●
- The collective changes it during each revolution, the cyclic changes all blades together
 - Both change it during each revolution, but only the cyclic changes all blades together
 - The cyclic changes it during each revolution, the collective changes all blades together
- 80.** A pilot pushes the cyclic stick forward to fly forward. Which chain of events takes place at the swash plate and the blades? ●
- The swash plate is lifted, the control rods move and the blade pitch rises on every blade at once
 - The swash plate turns, the bell cranks move and the blade pitch is held through the revolution
 - The swash plate tilts, the control rods move and the blade pitch varies through the revolution
- 81.** Explain why the rotor plane must be tilted when changing from a hover into forward flight, in terms of the law of acceleration ●
- A force is needed in the direction of flight, and thrust acts perpendicular to the rotor plane
 - A torque is needed in the direction of flight, and thrust acts perpendicular to the rotor mast
 - A force is needed in the direction of flight, and thrust acts parallel to the rotor plane
- 82.** In straight-ahead flight the blade has minimum incidence in front and maximum incidence at the rear. What is the result for the rotor plane? ●
- It tilts forward, giving lift and forward thrust
 - It tilts backward, giving lift and rearward thrust
 - It stays level, giving lift but no thrust
- 83.** Explain why raising the collective lever makes the rotor produce more lift ●
- The rotational speed of every blade rises, and with it the chord
 - The pitch angle of every blade rises, and with it the angle of attack
 - The plane of rotation of every blade rises, and with it the relative wind
- 84.** Identify which hinge allows the pitch change commanded from the cockpit and which allows the blade to move under aerodynamic forces ●
- The feathering hinge for the pitch change, the flapping hinge for the movement
 - The flapping hinge for the pitch change, the feathering hinge for the movement
 - The lead-lag hinge for the pitch change, the flapping hinge for the movement
- 85.** A pilot raises the collective lever to climb. The engine power changes and so does the torque effect on the fuselage. Which control must then vary the tail rotor thrust? ●
- The collective lever
 - The tail rotor pedals
 - The cyclic pitch stick
- 86.** Explain why varying the tail rotor thrust makes the helicopter turn around the rotor mast axis ●
- The thrust no longer exactly balances the torque reaction on the fuselage
 - The thrust no longer exactly balances the lift produced by the main rotor
 - The thrust tilts the main rotor disk away from the rotor mast axis

87. A pilot pushes the left pedal forward during a hover and holds the new heading. What else now needs attention? ●
- a. The pitch angle, because the collective lever has changed
 - b. The rotor RPM, because the tail rotor pitch has changed
 - c. The lateral drift, because the tail rotor thrust has changed

Answers — Cyclic, collective and anti-torque controls

59. c Lift counteracts weight. (L1, PDF p. 36)
60. b Autorotation allows a safe landing without engine power. (L1, PDF p. 36)
61. a Helicopters tilt the rotor blade plane instead of deflecting surfaces. (L1, PDF p. 38)
62. b Forward cyclic tilts the rotor plane forward. (L1, PDF p. 38)
63. a Collective up raises the pitch of all blades together. (L1, PDF p. 40)
64. b More pitch gives more lift and the helicopter climbs. (L1, PDF p. 40)
65. c Pedals turn the helicopter around the mast axis. (L1, PDF p. 42)
66. a Pedals change tail rotor blade pitch. (L1, PDF p. 42)
67. a Tilting the rotor plane lets the thrust act in any direction, which is how directional control is obtained. (L2, PDF p. 36)
68. a The rotor provides both lift and thrust. (L2, PDF p. 36)
69. a Tilting the rotor plane lets thrust act in any direction. (L2, PDF p. 36)
70. b The manual says it is fully controllable and the pilot selects the touchdown location. (L2, PDF p. 36)
71. c To point thrust forward, the plane must tilt, because thrust is perpendicular to it. (L2, PDF p. 38)
72. c Cyclic changes the incidence per revolution, giving different lift around the disc. (L2, PDF p. 38)
73. b The collective lever changes engine power as well as blade pitch, and with it the torque. (L2, PDF p. 40)
74. a Collective movement also changes engine power and torque effects. (L2, PDF p. 40)
75. b Pedal movement changes tail rotor blade pitch, which varies tail rotor thrust and yaws the aircraft. (L2, PDF p. 42)
76. a Tail rotor thrust is varied via blade pitch. (L2, PDF p. 42)
77. b A powered rotor doing all the work also produces the torque reaction that must be counteracted. (L3, PDF p. 36)
78. a Thrust acts perpendicular to the rotor plane, so tilting the plane sideways points the thrust sideways. (L3, PDF p. 36)
79. c Cyclic pitch varies around the disc each revolution while collective pitch moves all blades by the same amount. (L3, PDF p. 38)
80. c A cyclic input tilts the swash plate, which moves the control rods and varies blade pitch through each revolution. (L3, PDF p. 38)
81. a Accelerating forward needs a forward force, and the only way to point the thrust forward is to tilt the rotor plane. (L3, PDF p. 38)
82. a More lift at the rear and less at the front tilts the plane forward, producing lift and forward thrust. (L3, PDF p. 38)
83. b More pitch means more angle of attack on every blade, so the rotor makes more lift. (L3, PDF p. 40)
84. a Feathering hinges take the commanded pitch change, flapping hinges let the blade move under the air loads. (L3, PDF p. 40)
85. b Torque is compensated by the tail rotor, whose thrust the pilot varies with the pedals. (L3, PDF p. 40)
86. a The tail rotor thrust is what holds the torque reaction in balance, so changing it lets the fuselage turn. (L3, PDF p. 42)
87. c Tail rotor thrust is what produces the translating tendency, so changing it changes the drift the pilot must hold off. (L3, PDF p. 42)

Swash plate, dissymmetry of lift and flapping

PDF pages 44–54 • 44 questions

LEVEL 1 — RECALL

88. How is the collective pitch change achieved at the swash plate ●
 - a. By rotating the stationary swash plate
 - b. By lifting or lowering the swash plate
 - c. By tilting the swash plate to one side
89. How is the cyclic pitch change achieved at the swash plate ●
 - a. By tilting the swash plate
 - b. By lifting the whole swash plate
 - c. By turning the stationary part
90. In a hover with no wind, how does the lift of the rotor blades compare ●
 - a. The advancing blade generates more lift
 - b. The retreating blade generates more lift
 - c. All blades generate the same amount of lift
91. What is dissymmetry of lift ●
 - a. The difference in lift in and out of ground effect
 - b. The difference in lift between main and tail rotor
 - c. The difference in lift between advancing and retreating blade
92. Where does the blade's centre of gravity move when the blade flaps upwards ●
 - a. Closer towards the trailing edge
 - b. Closer towards the blade tip
 - c. Closer towards the rotational axis
93. Where does the blade's centre of gravity move when the blade flaps downwards ●
 - a. Closer to the blade tip
 - b. Closer to the rotational axis
 - c. Closer to the leading edge
94. What happens to the angle of attack as the advancing blade flaps up ●
 - a. It increases
 - b. It stays the same
 - c. It decreases
95. What happens to the angle of attack as the retreating blade flaps down ●
 - a. It stays the same
 - b. It increases
 - c. It decreases
96. Which two ways are given to achieve the flexibility for blade flapping ●
 - a. A lead-lag hinge or a larger and stiffer tail rotor
 - b. A flapping hinge or blade properties that absorb the forces
 - c. A feathering hinge or a stronger swash plate bearing

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 97.** Compare what acts on the stationary part of the swash plate with what the rotating part acts on ●
- The rotor head acts on it through a bearing; it drives the bell cranks through control rods
 - Control rods and bell cranks from the cockpit act on it; it drives the blades through control rods
 - Control rods from the blades act on it; it drives the cockpit controls through bell cranks
- 98.** Why does the swash plate have to act as an interface ●
- It transfers torque between the main rotor and the tail rotor drive system
 - It transfers control signals between the stationary input system and the rotating rotor
 - It transfers engine power between the main gearbox and the rotating rotor mast
- 99.** What is the direct result on the rotor blades when the swash plate moves? ●
- The blades move forward and backward on lead-lag hinges
 - The control rods move and the blade pitch angle changes
 - The bearing locks and the blades stop their flapping
- 100.** A helicopter is hovering with no wind at all. Why is there no dissymmetry of lift in this condition? ●
- There is no pitch angle on either blade, so both blades have the same parameters
 - There is no rotational speed at the blade tips, so both blades have the same parameters
 - There is no airstream from speed of flight, so both blades have the same parameters
- 101.** Why does the advancing blade produce more lift in forward flight ●
- The opposing airstream is added to its rotational speed
 - The opposing airstream is subtracted from its rotational speed
 - Its pitch angle is increased by the collective
- 102.** Why does the retreating blade produce less lift in forward flight ●
- Its pitch angle is reduced by the tail rotor pedals
 - The opposing airstream is subtracted from its rotational speed
 - The opposing airstream is added to its rotational speed
- 103.** Explain why an upward flapping rotor blade accelerates while the rotor RPM stays constant ●
- Its centre of gravity moves closer to the trailing edge
 - Its centre of gravity moves closer to the blade tip
 - Its centre of gravity moves closer to the rotational axis
- 104.** A mass point on a rotating disk is moved towards the edge of the disk. What happens to the rotational speed and why? ●
- It decelerates, because the mass resists the tangential acceleration
 - It stays constant, because the rotor RPM is fixed by the engine
 - It accelerates, because the mass tries to hold its higher speed
- 105.** A mass point on a rotating disk is moved towards the centre of rotation. What does it do to the rotational speed? ●
- It has no effect on the rotational speed
 - It tries to accelerate the rotational speed
 - It tries to decelerate the rotational speed
- 106.** With constant rotor RPM, what does an upward flapping blade do ●
- It accelerates and moves forward
 - It decelerates and moves backward
 - It accelerates and moves backward

- 107.** With constant rotor RPM, what does a downward flapping blade do ●
- It decelerates and moves forward
 - It decelerates and moves backward
 - It accelerates and moves forward
- 108.** Why is the installation of a lead-lag hinge required ●
- Flapping blades accelerate or decelerate and move forward or backward
 - Flapping blades change their pitch angle during each revolution
 - Flapping blades change the direction of the torque reaction
- 109.** Why does the advancing blade flap up ●
- It experiences increased relative wind and generates more lift
 - It experiences decreased relative wind and generates less lift
 - Its centre of gravity moves closer to the blade tip
- 110.** The retreating blade experiences a decreased relative wind. Which sequence follows according to the manual? ●
- Less lift, blade flaps down, angle of attack and lift increase
 - More lift, blade flaps up, angle of attack and lift decrease
 - Less lift, blade flaps up, angle of attack increases
- 111.** The advancing blade flaps up in forward flight. What finally happens to its lift? ●
- The lift decreases as the angle of attack decreases
 - The lift stays constant as the pitch is constant
 - The lift increases as the angle of attack increases
- 112.** Is blade flapping alone sufficient to restore rotor balance ●
- No, it helps to balance but is not sufficient
 - Yes, it fully restores rotor balance
 - No, it increases the dissymmetry of lift
- 113.** Why must a rotor blade be able to move and occupy a new position ●
- Because the swash plate rotates with the rotor
 - Because aerodynamic forces act on the rotor blade
 - Because the torque reaction turns the fuselage
- 114.** Blade flapping compensates dissymmetry of lift. Which hinge lets the blade move up and down for this? ●
- Feathering hinge
 - Lead-lag hinge
 - Flapping hinge
- 115.** A pilot makes a cyclic control input. Which chain of effects follows according to the manual? ●
- The tail rotor thrust changes, which generates torque
 - The swash plate lifts, which generates ground effect
 - The rotor plane tilts, which generates a lead-lag movement

LEVEL 3 — APPLY AND REASON

- 116.** Explain why the two parts of the swash plate are held parallel but joined through a bearing ●
- So the cockpit input can be passed to blades that are turning with the rotor
 - So the rotor plane can be tilted without moving the control rods above it
 - So the cockpit input can be passed to blades that are fixed to the rotor mast

- 117.** During a ground check the whole swash plate is seen to rise without tilting, and every blade takes a larger pitch angle. What will the helicopter do if this is done in flight? ●
- It yaws left, because the tail rotor pitch is increased
 - It flies forward, because the rotor plane is tilted forward
 - It climbs, because all blades gain pitch together
- 118.** Name the cockpit control that changes a blade pitch angle without acting through the main rotor swash plate ●
- The cyclic stick, which changes the main rotor blade pitch
 - The pedals, which change the tail rotor blade pitch
 - The collective lever, which changes the main rotor blade pitch
- 119.** A technician watches the swash plate while a helper moves one cockpit control, and the whole swash plate moves up without tilting. Which control is being moved? ●
- The cyclic stick
 - The collective lever
 - The tail pedals
- 120.** During a control check the swash plate tilts but its mean height does not change. Which pitch change is being made? ●
- Collective pitch change
 - Tail rotor pitch change
 - Cyclic pitch change
- 121.** State where on the rotor disc the highest airspeed occurs when a helicopter flies forward ●
- At the tip of the advancing blade
 - At the root of the advancing blade
 - At the tip of the retreating blade
- 122.** A helicopter hovers on the spot in a steady strong wind. Why can its blades not all produce the same lift? ●
- The wind raises the pitch angle on one half of the disc and lowers it on the other
 - The wind increases the rotational speed of the whole rotor disc
 - The wind adds to the speed of one half of the disc and subtracts from the other
- 123.** A blade flaps up and therefore accelerates and moves forward in the plane of rotation. Which hinge has to allow this? ●
- The lead-lag hinge
 - The flapping hinge
 - The feathering hinge
- 124.** In forward flight the advancing blade flaps up. Combining the flapping and centre-of-mass facts, how does this blade move in the plane of rotation?
- It accelerates and moves forward (leads)
 - It decelerates and moves backward (lags)
 - It keeps its position in the plane
- 125.** In forward flight the retreating blade flaps down. Combining the flapping and centre-of-mass facts, how does this blade move in the plane of rotation? ●
- It decelerates and moves backward (lags)
 - It keeps its position in the plane
 - It accelerates and moves forward (leads)

- 126.** Explain why blade flapping is described as a self-correcting effect ●
- The blade with more lift flaps down, which reduces its angle of attack and its lift
 - The blade with more lift flaps up, which reduces its pitch angle and its lift
 - The blade with more lift flaps up, which reduces its angle of attack and its lift
- 127.** Identify on which blade flapping increases the angle of attack, and give the reason it has to ●
- The advancing blade, because it is the one producing too little lift
 - The retreating blade, because it is the one producing too much lift
 - The retreating blade, because it is the one producing too little lift
- 128.** Why does upward flapping reduce the angle of attack of the advancing blade, using the relative wind rule? ●
- Moving up reduces the blade pitch angle
 - Moving up adds an upward relative wind component
 - Moving up adds a downward relative wind component
- 129.** A rotor head is built with no flapping hinges at all. How can dissymmetry of lift still be balanced on this rotor? ●
- The feathering hinges let the blades change pitch instead
 - The lead-lag hinges let the blades move up and down instead
 - The blades themselves flex and absorb the forces acting on them
- 130.** A helicopter is in steady forward flight and the pilot makes no control input at all. Is there still lead-lag movement, and why? ●
- Yes, because dissymmetry of lift keeps the blades flapping
 - No, because lead-lag only follows a cyclic control input
 - Yes, because the collective lever holds the blades at a fixed pitch
- 131.** A pilot in forward flight moves the cyclic stick to the right. Which blade movement is produced in addition to the pitch change? ●
- A feathering movement, because the rotor plane tilts
 - A lead-lag movement, because the rotor RPM changes
 - A lead-lag movement, because the rotor plane tilts

Answers — Swash plate, dissymmetry of lift and flapping

- 88. b** Collective raises or lowers the swash plate. (L1, PDF p. 44)
- 89. a** Cyclic tilts the swash plate. (L1, PDF p. 44)
- 90. c** In a no-wind hover all blades have the same parameters. (L1, PDF p. 46)
- 91. c** The lift difference between advancing and retreating blade is dissymmetry of lift. (L1, PDF p. 46)
- 92. c** Upward flapping brings the CG closer to the axis. (L1, PDF p. 48)
- 93. a** Downward flapping moves the CG outward toward the tip. (L1, PDF p. 48)
- 94. c** Upward flapping decreases the angle of attack. (L1, PDF p. 50)
- 95. b** Downward flapping increases the angle of attack. (L1, PDF p. 50)
- 96. b** Either a flapping hinge or blades that absorb the forces. (L1, PDF p. 52)
- 97. b** The cockpit acts on the stationary part and the rotating part is connected onward to the blades. (L2, PDF p. 44)
- 98. b** Cockpit inputs are stationary but the blades rotate, so the swash plate links the two. (L2, PDF p. 44)
- 99. b** Swash plate motion moves control rods, which change pitch. (L2, PDF p. 44)
- 100. c** Without the airstream caused by the speed of flight there is nothing to add to one blade and subtract from the other. (L2, PDF p. 46)

- 101. a** The airstream adds to its speed, giving more lift. (L2, PDF p. 46)
- 102. b** The airstream subtracts from its speed, giving less lift. (L2, PDF p. 46)
- 103. c** A mass moved towards the centre of rotation tries to speed the rotation up, so the up-flapping blade accelerates. (L2, PDF p. 48)
- 104. a** Moving outward needs higher circumferential speed; inertia resists and slows the rotation. (L2, PDF p. 48)
- 105. b** Moving inward, it keeps its higher speed and so accelerates the rotation. (L2, PDF p. 48)
- 106. a** Its CG moves inward, so it speeds up and leads. (L2, PDF p. 48)
- 107. b** Its CG moves outward, so it slows down and lags. (L2, PDF p. 48)
- 108. a** The forward and backward blade movement caused by flapping must be allowed by a lead-lag hinge. (L2, PDF p. 48)
- 109. a** More relative wind means more lift, so it flaps up. (L2, PDF p. 50)
- 110. a** The retreating blade flaps down, which raises angle of attack and lift. (L2, PDF p. 50)
- 111. a** Upward flapping reduces angle of attack and so lift. (L2, PDF p. 50)
- 112. a** Flapping helps but does not fully restore balance. (L2, PDF p. 50)
- 113. b** Aerodynamic forces require the blade to move. (L2, PDF p. 52)
- 114. c** Flapping hinges allow the up-down movement. (L2, PDF p. 52)
- 115. c** Cyclic tilts the plane, and that generates lead-lag. (L2, PDF p. 54)
- 116. a** The bearing lets the upper part turn with the rotor while the parallel lower part stays still, passing the input across. (L3, PDF p. 44)
- 117. c** Lifting the swash plate is a collective pitch increase, which makes more lift and a climb. (L3, PDF p. 44)
- 118. b** The swash plate serves the main rotor blades, while the pedals act on the tail rotor blades. (L3, PDF p. 44)
- 119. b** Lifting the swash plate is collective; tilting would be cyclic. (L3, PDF p. 44)
- 120. c** Tilting the swash plate produces cyclic pitch change. (L3, PDF p. 44)
- 121. a** Speed rises towards the tip and the opposing airstream adds on the advancing side, so the advancing tip is fastest. (L3, PDF p. 46)
- 122. c** Wind is a relative wind just like forward speed, so it adds on the advancing side and subtracts on the retreating side. (L3, PDF p. 46)
- 123. a** Movement forward and backward in the plane of rotation is taken by the lead-lag hinge. (L3, PDF p. 48)
- 124. a** The advancing blade flaps up, its CG moves inward, so it accelerates and moves forward. (L3, PDF p. 48)
- 125. a** The retreating blade flaps down, its CG moves outward, so it decelerates and lags. (L3, PDF p. 48)
- 126. c** The extra lift causes the up-flap, and the up-flap takes the angle of attack and the lift back down again. (L3, PDF p. 50)
- 127. c** The retreating blade is short of lift, and flapping down raises its angle of attack and so its lift. (L3, PDF p. 50)
- 128. c** Relative wind is opposite to blade motion, so an upward-moving blade meets air from above, reducing the angle of attack. (L3, PDF p. 50)
- 129. c** The manual allows blade properties that absorb the forces as the alternative to a flapping hinge. (L3, PDF p. 52)
- 130. a** Dissymmetry of lift makes the blades flap in forward flight even with no input, and flapping always makes lead-lag. (L3, PDF p. 54)
- 131. c** A cyclic input tilts the rotor plane and the tilt generates a lead-lag movement. (L3, PDF p. 54)

Translating tendency, ground effect and autorotation

PDF pages 56–60 • 29 questions

LEVEL 1 — RECALL

132. Which turning force does a counterclockwise turning main rotor generate on the fuselage ●
- A nose-down pitching force
 - A counterclockwise turning force
 - A clockwise turning force
133. During hover flight, which two jobs does the tail rotor thrust perform ●
- Compensate main rotor torque and maintain heading control
 - Maintain heading control and tilt the rotor disk
 - Generate lift and compensate main rotor torque
134. Within which height is a helicopter operating in ground effect ●
- Within two rotor diameters
 - Within one rotor diameter
 - Within half a rotor diameter
135. How does the power required to hover in ground effect compare with hovering out of ground effect? ●
- More power is required in ground effect
 - The same power is required in both
 - Less power is required in ground effect
136. During powered flight, how is the rotor drag overcome ●
- With rotor inertia
 - With engine power
 - With upward airflow
137. In which two failure cases must some other force sustain the rotor RPM ●
- Engine failure or broken drive shafts
 - Swash plate failure or broken control rods
 - Tail rotor failure or broken pedals

LEVEL 2 — UNDERSTAND AND EXPLAIN

138. A helicopter is in a steady hover. Which two jobs is the thrust of the tail rotor doing at the same time? ●
- Compensating the main rotor torque and preventing the lateral drift
 - Maintaining heading control and tilting the main rotor disk slightly
 - Compensating the main rotor torque and maintaining heading control
139. A helicopter in hover drifts sideways although torque is compensated. What causes this drift? ●
- Gyroscopic precession
 - The thrust of the tail rotor
 - Dissymmetry of lift
140. Which rotor thrust both compensates torque and causes translating tendency ●
- The total rotor thrust
 - The tail rotor thrust
 - The main rotor thrust

- 141.** Explain why ground effect becomes more pronounced the nearer the ground is approached ●
- Because it is caused by the surface interfering with the rotor airflow pattern
 - Because it is caused by the surface increasing the pitch angle of the blades
 - Because it is caused by the surface reducing the rotational speed of the blades
- 142.** What causes ground effect
- Increased air density close to the ground surface
 - Interference of the surface with the rotor airflow pattern
 - Interference of the tail rotor with the main rotor airflow
- 143.** A helicopter hovering out of ground effect descends into an in-ground effect hover. What happens to blade efficiency and why? ●
- It increases, because the reduced downward airflow increases the angle of attack
 - It decreases, because the increased downward airflow reduces the angle of attack
 - It increases, because the pilot must increase the pitch angle of the blades
- 144.** Explain why a helicopter in autorotation has to descend ●
- Because the engine power is needed to overcome the drag of the rotor blades
 - Because the collective pitch is lowered to its minimum stop during autorotation
 - Because the airflow up through the rotor in the descent supplies the energy
- 145.** What provides the energy to overcome blade drag and turn the rotor during autorotation
- The tail rotor thrust through the gearbox
 - The engine power through the drive shafts
 - The airflow passing up through the rotor during descent
- 146.** As altitude decreases in autorotation, what happens to the energy stored in the turning rotor? ●
- More potential energy is converted into kinetic energy stored in the rotor
 - More kinetic energy is converted into potential energy of the helicopter
 - The stored rotor energy is lost to drag and the rotor comes to a stop
- 147.** Why must the rotor RPM be sustained after an engine failure ●
- So that a controlled flight can be continued to the ground
 - So that the tail rotor can be stopped during the descent
 - So that the engine can be restarted during the descent

LEVEL 3 — APPLY AND REASON

- 148.** Name the cockpit control the pilot uses to hold off the drift caused by translating tendency ●
- The cyclic stick, which tilts the main rotor disk
 - The pedals, which tilt the tail rotor thrust line
 - The collective lever, which lifts the main rotor disk
- 149.** Explain how tilting the main rotor disk stops the drift caused by the tail rotor ●
- The lift then acts partly against the torque direction
 - The thrust then acts partly against the rotor mast axis
 - The thrust then acts partly in the opposite sideways direction
- 150.** During hover, tail rotor thrust makes the helicopter drift to the right. How must the main rotor disk be tilted to prevent this? ●
- Slightly to the right
 - Slightly forward
 - Slightly to the left

151. Explain why the reduced downward airflow in ground effect increases the angle of attack of the blades ●
- The relative wind direction changes, and the angle of attack is measured from it
 - The plane of rotation changes, and the angle of attack is measured from it
 - The pitch angle changes, and the angle of attack is measured from it
152. A helicopter settles into a hover close to the ground and the pilot wants to stay there. Which lever does he move and what happens to the power needed? ●
- He lowers the collective, and more power is needed
 - He lowers the collective, and less power is needed
 - He raises the collective, and more power is needed
153. A helicopter hovering just above the ground climbs slowly until it is just under one rotor diameter up. What has happened to ground effect? ●
- It is stronger and still present
 - It is lost once the skids leave the ground
 - It is weaker but still present
154. In ground effect the lift vector of the blades becomes more vertical. Why does that reduce the power the pilot needs? ●
- More of the rotor force acts straight along the blade span
 - More of the rotor force acts straight against the weight
 - More of the rotor force acts straight against the drag
155. A pilot descends from an out-of-ground effect hover into ground effect and wants to remain in the ground effect hover. What must he do and why? ●
- Increase the blade pitch angle, because rotor efficiency decreases
 - Keep the blade pitch angle, because efficiency is unchanged
 - Reduce the blade pitch angle, because rotor efficiency increases
156. A helicopter hovers in ground effect and the pilot climbs out of ground effect without changing the collective. Based on the text, what happens to the power requirement to keep hovering? ●
- It decreases
 - It stays the same
 - It increases
157. State what the collective pitch controls once the engine has failed ●
- The engine power that keeps the rotor turning in the descent
 - The force that keeps the rotor turning in the descent
 - The tail rotor thrust that keeps the heading in the descent
158. An engine fails in cruise and the pilot chooses a field to land in. What makes that choice possible? ●
- The rotor keeps turning at an RPM that provides aircraft control
 - The engine keeps turning at an RPM that provides hydraulic power
 - The rotor keeps turning at an RPM that provides enough lift to hover
159. Explain why the kinetic energy stored in the turning rotor must not all be spent during the autorotative descent ●
- It is needed to restart the engine before the touchdown
 - It is needed to cushion the touchdown near the ground
 - It is needed to drive the tail rotor during the touchdown

- 160.** An engine fails in cruise. In which order does the pilot use energy during the autorotation? ●
- Rotor kinetic energy cushions the descent, then altitude drives the rotor at touchdown
 - Altitude drives the rotor in the descent, then rotor kinetic energy cushions touchdown
 - Engine power drives the rotor in the descent, then altitude cushions touchdown

Answers — Translating tendency, ground effect and autorotation

- 132. c** The fuselage force is opposite, so clockwise. (L1, PDF p. 56)
- 133. a** Tail rotor thrust compensates torque and maintains heading. (L1, PDF p. 56)
- 134. b** In-ground effect is within one rotor diameter. (L1, PDF p. 58)
- 135. c** Hovering in ground effect needs less power. (L1, PDF p. 58)
- 136. b** In powered flight, engine power overcomes rotor drag. (L1, PDF p. 60)
- 137. a** Engine failure or broken drive shafts require another force. (L1, PDF p. 60)
- 138. c** In the hover the tail rotor thrust both compensates the main rotor torque and maintains heading control; the drift it causes is held off with the main rotor disk instead. (L2, PDF p. 56)
- 139. b** The tail rotor's sideways thrust causes translating tendency. (L2, PDF p. 56)
- 140. b** The same tail rotor thrust compensates torque and causes the drift. (L2, PDF p. 56)
- 141. a** The effect comes from the surface interfering with the rotor airflow, so it grows as the surface gets closer. (L2, PDF p. 58)
- 142. b** The ground surface interferes with the rotor airflow. (L2, PDF p. 58)
- 143. a** Reduced downwash raises angle of attack and efficiency. (L2, PDF p. 58)
- 144. c** The upward airflow that keeps the rotor turning only exists while the helicopter is descending. (L2, PDF p. 60)
- 145. c** Upward airflow during descent drives the rotor. (L2, PDF p. 60)
- 146. a** Descent converts potential into kinetic energy stored in the rotor. (L2, PDF p. 60)
- 147. a** Rotor RPM keeps the helicopter controllable to the ground. (L2, PDF p. 60)
- 148. a** Tilting the rotor disk is a cyclic input, so the cyclic stick is the control that holds off the drift. (L3, PDF p. 56)
- 149. c** Thrust acts perpendicular to the rotor plane, so tilting the disk the other way points some thrust against the drift. (L3, PDF p. 56)
- 150. c** The disk is tilted opposite to the drift, so to the left for a right drift. (L3, PDF p. 56)
- 151. a** A slower downward airflow changes the relative wind, and the angle of attack is taken from the relative wind. (L3, PDF p. 58)
- 152. b** Reducing blade pitch is done with the collective, and hovering in ground effect needs less power. (L3, PDF p. 58)
- 153. c** Ground effect reaches to one rotor diameter and weakens with height, so it is still there but reduced. (L3, PDF p. 58)
- 154. b** Lift is what overcomes weight, so a more vertical lift vector puts more of the rotor force where it is needed. (L3, PDF p. 58)
- 155. c** Entering ground effect raises angle of attack and efficiency, so pitch is reduced to stay in the hover. (L3, PDF p. 58)
- 156. c** Hovering out of ground effect needs higher power than in ground effect. (L3, PDF p. 58)
- 157. b** With no engine the collective no longer sets engine power; it controls the force that sustains rotor RPM. (L3, PDF p. 60)
- 158. a** Autorotation keeps the rotor at a controlling RPM, so the helicopter stays fully controllable and the pilot picks the spot. (L3, PDF p. 60)
- 159. b** The stored rotor energy is what the pilot uses in the second phase to cushion the touchdown. (L3, PDF p. 60)
- 160. b** Phase 1 converts altitude into rotor energy; phase 2 uses the rotor energy for touchdown. (L3, PDF p. 60)